K-Nearest Neighbors (KNN): Can you answer these 20 Questions?

Machine Learning Interview Questions Part 21

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Shahidullah Kawsar Following

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How to train a ML model using KNN in 5 steps:

1. Basic Idea (Similarity-Based Learning): K-Nearest Neighbors (KNN) predicts the output of a new data point by looking at the K most similar data points in the training dataset. The assumption is that similar inputs produce similar outputs, so nearby observations are likely to belong to the same class or have similar values.

2. Distance Calculation (Finding Neighbors): To determine similarity, KNN calculates the distance between the new data point and all training samples using metrics such as Euclidean distance, Manhattan distance, or cosine similarity. The algorithm then selects the K closest points based on these distance measurements.

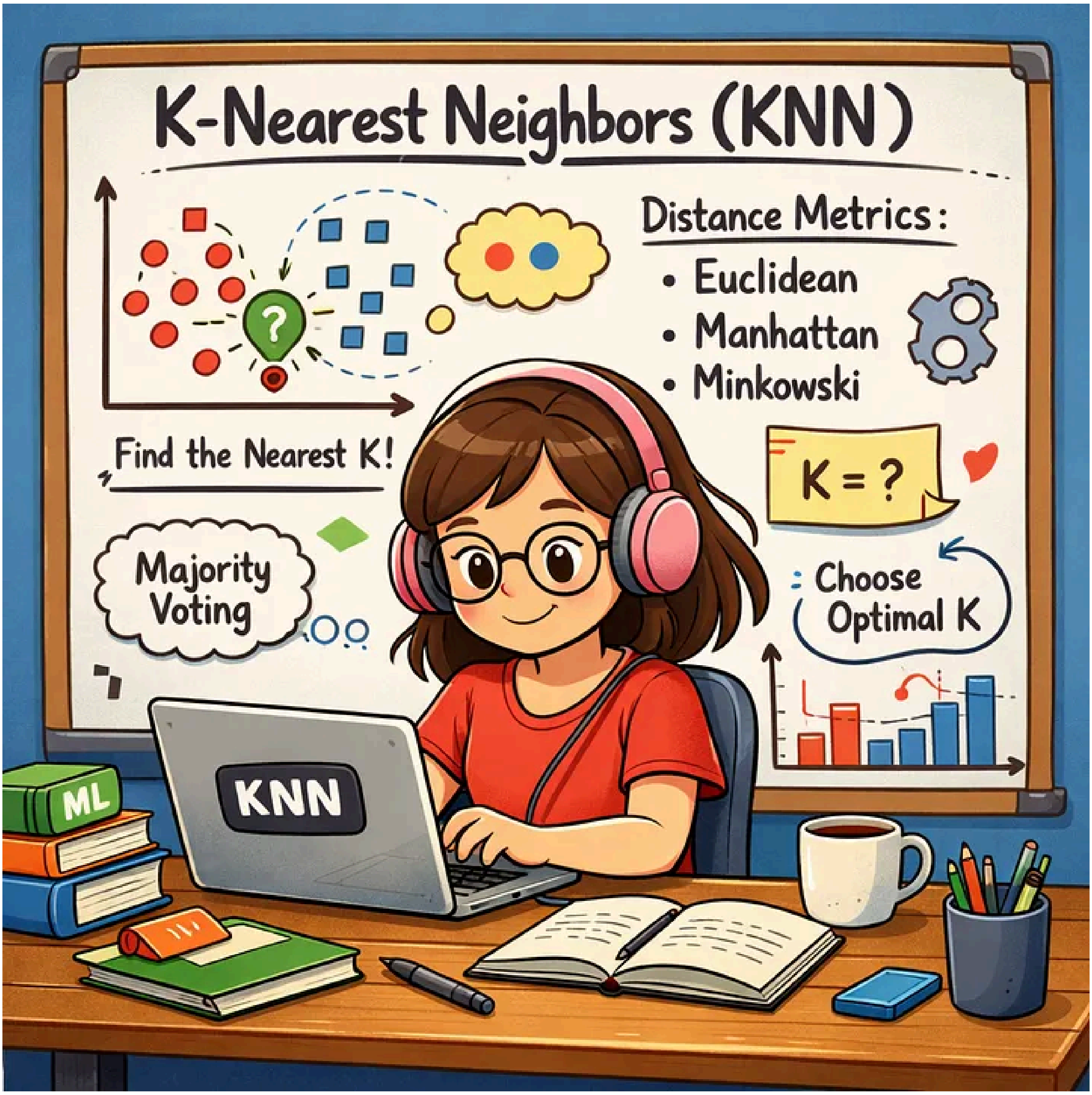
3. Prediction Rule (Voting or Averaging): Once the K nearest neighbors are identified, the algorithm makes a prediction.

- For classification, it uses majority voting among neighbors.
- For regression, it computes the average of the neighbors' values.

4. No Training Phase (Lazy Learning): KNN is called a lazy learning algorithm because it does not build a model during training. Instead, it simply stores the dataset and performs computations only when a prediction is required, which makes training fast but prediction slower.

5. Key Considerations (K Value and Scaling): The choice of K strongly affects performance: small K may cause overfitting, while large K may over smooth predictions. Additionally, feature scaling (normalization or standardization) is important because KNN relies on distance calculations, and features with larger ranges can dominate the results.

Let's check your basic knowledge of KNN. Here are 20 questions for your next interview.



Source: This image is generated by ChatGPT

1. What is the core principle behind KNN?

(A) Optimization-based learning
(B) Similarity-based learning
(C) Probabilistic modeling
(D) Gradient descent

2. Why is KNN considered a lazy learner?

(A) It delays predictions

(B) It stores data without training

(C) It ignores features

(D) It uses random sampling

3. In KNN, What does the parameter K represent?

(A) Number of features

(B) Number of clusters

(C) Number of nearest neighbors

(D) Number of iterations

4. In KNN, What happens when K = 1?

(A) High bias

(B) High variance

(C) Underfitting

(D) No prediction

5. Which distance metric is commonly used in KNN?

(A) Cross-entropy

(B) Euclidean distance

(C) Gradient norm

(D) KL divergence

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6. Why is feature scaling important in KNN?

(A) Improves convergence

(B) Prevents overfitting

(C) Equalizes feature influence

(D) Reduces memory

7. How is prediction done in classification using KNN?

(A) Mean value
(B) Majority voting
(C) Gradient descent
(D) Random selection

8. What is the bias-variance tradeoff in KNN?

(A) Small K → high bias
(B) Large K → high variance
(C) Small K → high variance
(D) No tradeoff

9. What is a major disadvantage of KNN?

(A) High training cost
(B) High prediction cost
(C) Requires labels
(D) Cannot classify

10. What is weighted KNN?

(A) Random weighting
(B) Equal weighting
(C) Distance-based weighting
(D) Feature weighting

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11. What problem arises in high-dimensional KNN?

(A) Overfitting

(B) Curse of dimensionality

(C) Underfitting

(D) Memory leak

12. What is the role of KD-Trees in KNN?

(A) Reduce memory

(B) Speed up neighbor search

(C) Improve accuracy

(D) Reduce bias

13. In KNN, what happens if K equals dataset size?

(A) Overfitting

(B) Underfitting

(C) Perfect accuracy

(D) No prediction

14. Why is KNN non-parametric?

(A) Uses parameters

(B) No fixed functional form

(C) Uses gradients

(D) Uses clusters

15. Why does KNN struggle with irrelevant features?

(A) Reduces speed

(B) Distorts distance

(C) Causes bias

(D) Removes neighbors

16. **What is KNN imputation?**

(A) Removing data

(B) Filling missing values using neighbors

(C) Random fill

(D) Mean fill only

17. **What is the prediction complexity of KNN?**

(A) O(1)

(B) O(log n)

(C) O(n)

(D) $O(n^2)$

18. **What is the difference between KNN and K-means?**

(A) Both supervised

(B) KNN supervised, K-means unsupervised

(C) Both unsupervised

(D) Same algorithm

19. **Why does KNN not scale well?**

(A) High training cost

(B) High prediction latency

(C) Low accuracy

(D) No features

20. **Why can KNN overfit despite no training phase?**

(A) Large K

(B) Small K sensitivity to noise

(C) No features

(D) Random predictions

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Happy learning! Feel free to discuss your thoughts on these questions in the comment section. Don't forget to **clap** and **share** the quiz link with your friends & LinkedIn connections.

Reference: [1] MCQ-based ML Interview Prep Questions [**Link**] and solutions [**Link**]

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